

PREM KUMAR VEPANAJERI

Data Analyst

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SUMMARY

Experienced Data Analyst with 2+ years of hands-on experience in insurance, financial services, and cloud environments, leveraging SQL, Python, R, and advanced analytics tools to improve policy performance and optimize claims processing. Proficient in developing predictive models, interactive dashboards, and ETL pipelines, while ensuring compliance with NAIC, PCI-DSS, and HIPAA regulations. Strong team player with a track record of collaborating with cross-functional teams to integrate and streamline data across multiple platforms like AWS, Hadoop, and Redshift. Adept at using Power BI, Tableau, and machine learning models to deliver actionable insights and drive data-driven decisions for improved operational efficiency and fraud detection accuracy.

SKILLS

- Methodologies:** SDLC, Agile, Waterfall, Scrum
- Programming Languages:** SQL, Python, R
- Packages & Libraries:** NumPy, Pandas, Matplotlib, Scipy, ggplot2, Seaborn, Plotly, scikit-learn, TensorFlow, PySpark
- Visualization Tools:** Tableau, Power BI, MS Excel
- Databases:** MySQL, SQL Server, PostgreSQL, Oracle, MongoDB, Amazon Redshift
- Data Privacy & Compliance:** HIPAA, HL7
- ETL & Data Integration Tools:** Hadoop, SSIS, Talend, Apache NiFi, Informatica, Alteryx
- CRM & Analytics Platforms:** Salesforce, SAP, SAS, Google Analytics, HubSpot

EXPERIENCE

Progressive Insurance, CT, USA01/2025 - Present

Data Analyst

- Analyzed Insurance Data to improve policy profitability by tracking claims trends, underwriting performance, and the impact of pricing strategies within Progressive Insurance’s Risk & Claims Analytics Platform, using SQL and Python.
- Built Predictive Models using historical claims and policyholder data to forecast high-risk claims and potential policy lapses, enabling proactive interventions that reduced claim payouts by 4% and improved retention rates by 6%.
- Developed Interactive Dashboards in Tableau and Power BI to visualize KPIs such as claim settlement times, fraud detection rates, and policy conversion metrics, shortening reporting cycles by 20% and accelerating business decisions.
- Enhanced Fraud Detection in Claims Processing by analyzing large-scale insurance claims using machine learning models in Python and Scikit-learn, increasing fraud identification accuracy by 15% while reducing investigation time by 10%.
- Optimized Claims Data Processing using SQL and AWS Redshift, improving query performance and enabling faster analysis of large policy and claims datasets.
- Ensured Compliance with insurance regulations such as NAIC and state-level mandates while managing sensitive policyholder data, maintaining a 100% compliance audit record.
- Collaborated with Cross-Functional Teams including underwriting, claims, and actuarial departments to integrate data from multiple insurance systems, ensuring 99% data accuracy and streamlining reporting processes using AWS and Hadoop.

Sophos, Bangalore, India10/2021 – 09/2022

Data Analyst

- Applied DAX functions to create interactive and insightful dashboards in Power BI, which enhanced data visualization capabilities and improved stakeholder engagement by 20%.
- Designed and maintained ETL pipelines with AWS Glue and Apache Airflow, enhancing data integration processes and achieving an 18% increase in ETL performance.
- Experience in working with NoSQL databases and big data tools, including Hadoop, Hive, and Spark.
- Used SAS for advanced statistical analysis, data cleaning, and predictive modeling to identify trends and drive data-driven strategies, contributing to a 12% improvement in forecasting accuracy.
- Developed SQL scripts, views, triggers, and stored procedures to update and maintain databases, to support operations.
- Leveraged Salesforce for data analysis, creating custom reports, dashboards, and extracting actionable insights to support business decisions.
- Utilized SSIS and Talend Open Studio for ETL tasks, delivering technical solutions to business stakeholders.
- Executed SQL queries to access and analyse data from the relational database for data modeling and reporting.
- Achieved an 8% increase in operational efficiency and a reduction in delivery time by spearheading an end-to-end supply chain project using SQL, Spark, and GCP.
- Designed database schemas, tables, and views, and migrated financial data from multiple sources to the SQL Server database, ensuring 99% data accuracy and accessibility.

EDUCATION

Sacred Heart University, Fairfield CT01/2024 – 05/2025

Masters in computer science & information and technology

KL University09/ 2018 – 04/2022

Bachelors in mechanical engineering